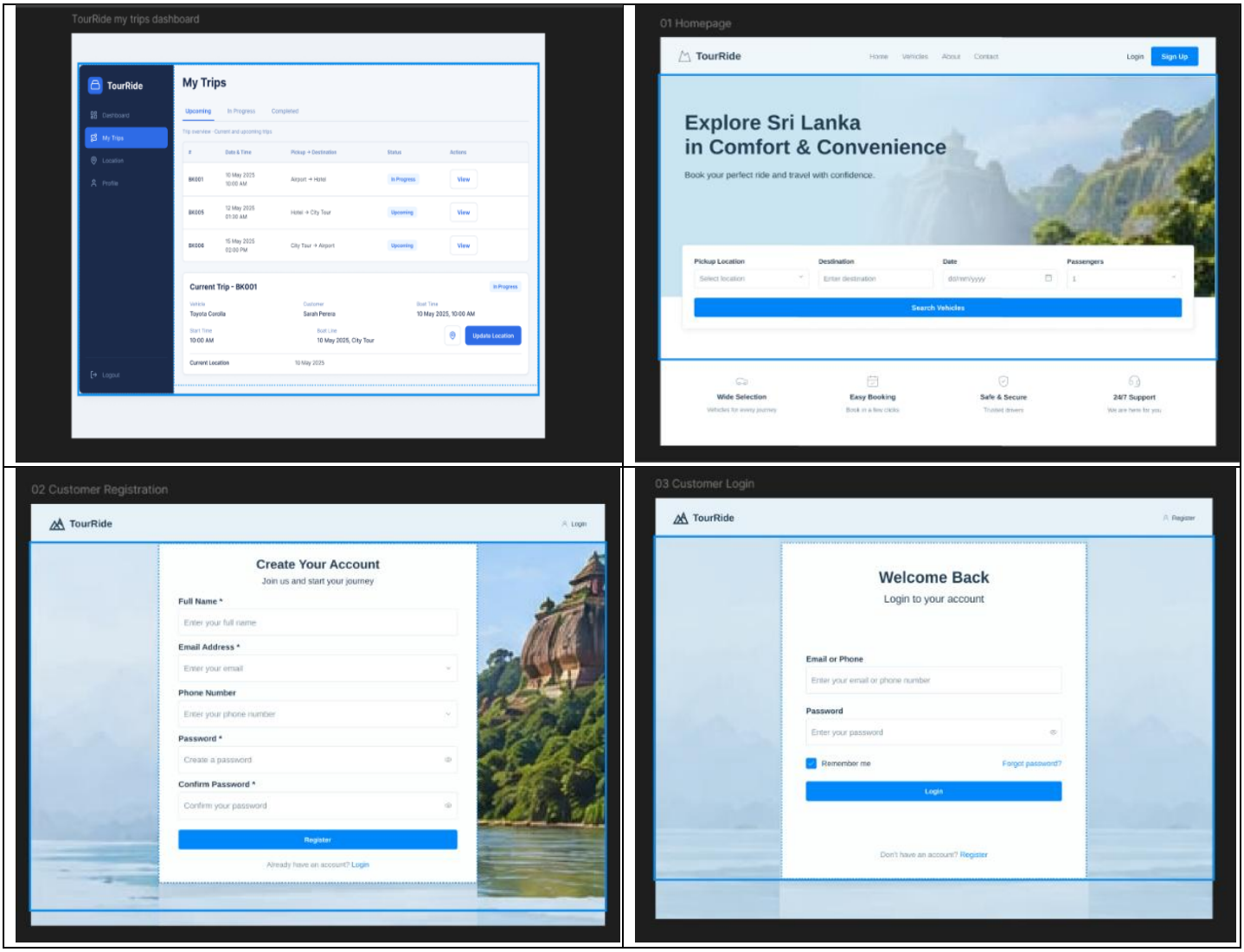
UI FORM VALIDATION

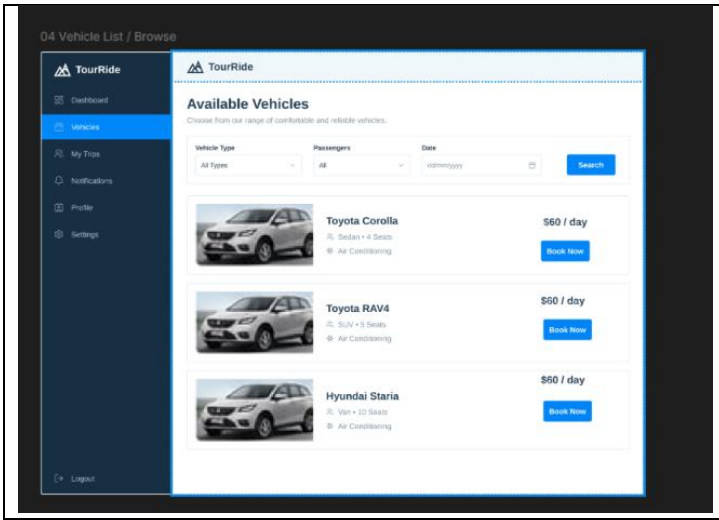
Tour Ride-Tourism Vehicle Booking and Trip Tracking System **24550825**

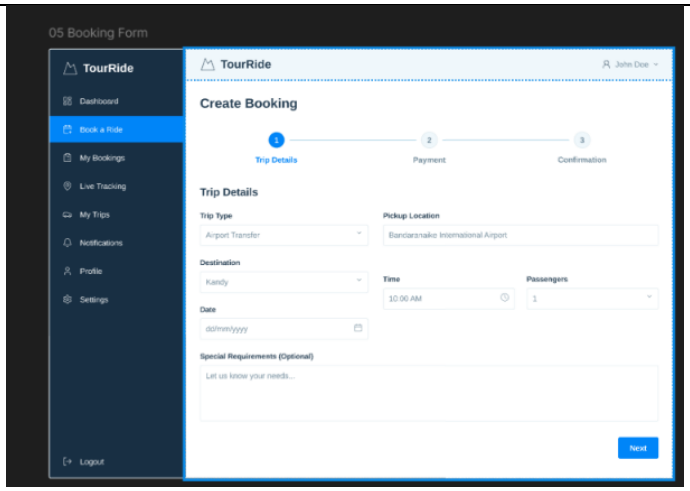
Purpose.

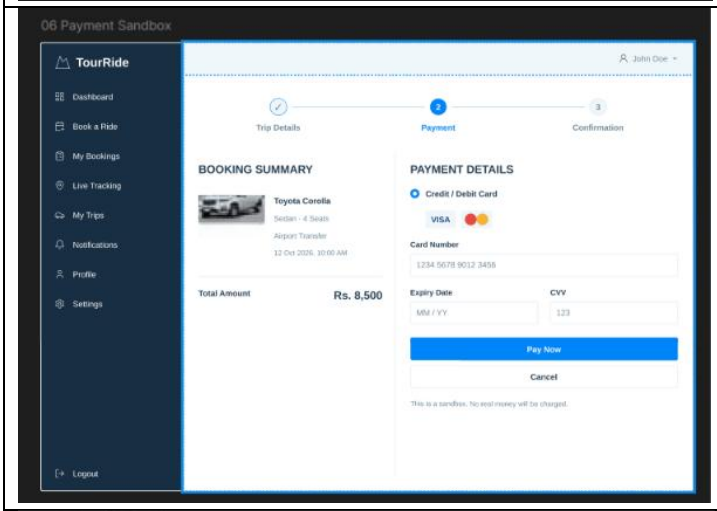
This section defines input-validation rules for the proposed user interface, based on Chapter 1 requirements, use cases and business rules. These are expected behaviours; describe them as implemented only after coding and testing.

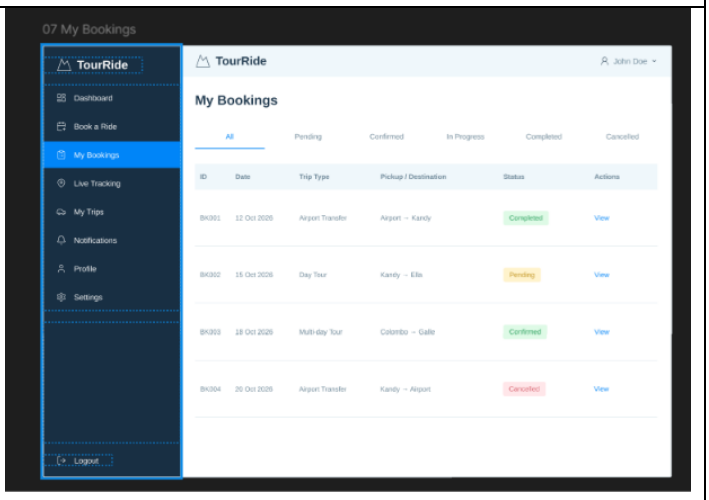
UI Wireframes

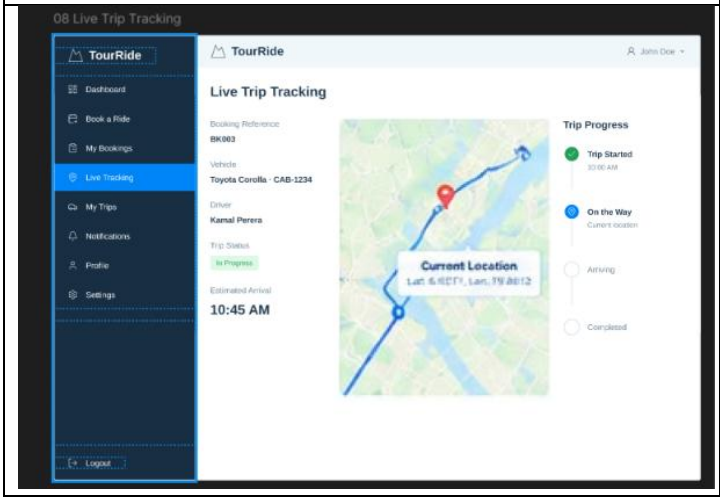


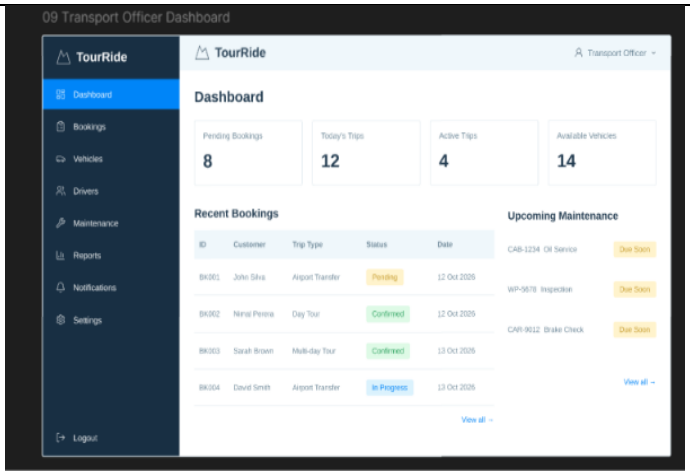

04 Vehicle List / Browse
TourRide
Available Vehicles
Toyota Corolla
$60 / day
Book Now
Toyota RAV4
$60 / day
Book Now
Hyundai Staria
$60 / day
Book Now


05 Booking Form
TourRide
Create Booking
Trip Details
Payment
Confirmation
Trip Details
Next


06 Payment Sandbox
TourRide
BOOKING SUMMARY
PAYMENT DETAILS
Credit / Debit Card
Total Amount
Rs. 8,500
Pay Now
Cancel


07 My Bookings
TourRide
My Bookings


08 Live Trip Tracking
TourRide
Live Trip Tracking
Trip Progress
Trip Started
On the Way
Arriving
Completed
Current Location
10:45 AM


09 Transport Officer Dashboard
TourRide
Dashboard
Pending Bookings
8
Today's Trips
12
Active Trips
4
Available Vehicles
14
Recent Bookings
Upcoming Maintenance

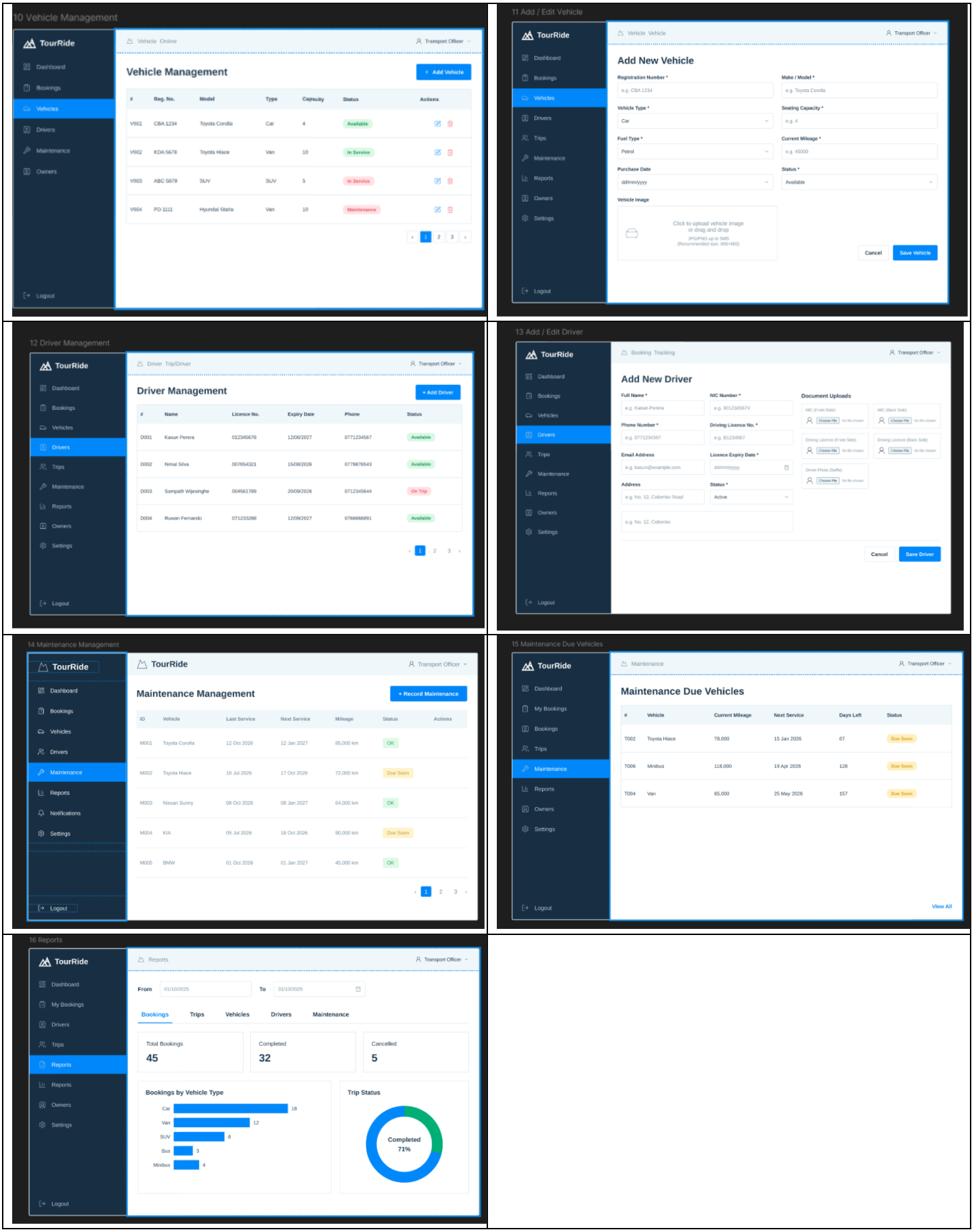


Table 1 – TourRide UI Designs

Validation Rules

Validate data on form submission and, where useful, while it is entered. Mark required fields clearly, show specific error messages and prevent invalid data from being saved.

Form / area	Validation requirements
Registration and Login	Required registration fields must not be empty. Validate email and phone formats; password confirmation must match and the chosen password policy must be met. Login requires credentials and role-based access. Error messages must not disclose whether a particular account exists.
Vehicle Booking	Pickup, destination, date, time, passenger count and trip type are required. Reject invalid or past date/time values. Passenger count must be a positive whole number; pickup and destination should differ. Reject bookings/assignments if capacity is insufficient, the vehicle is booked for an overlapping period or maintenance blocks its use.
Sandbox/Demo Payment	If included in the prototype, validate required test-form fields and formats and display the sandbox result clearly. Use test details only; do not claim a real payment was processed or store raw card security codes. Chapter 1 specifies demo payment only.
Vehicle Management	Validate required details such as registration number, make/model, year and capacity. Capacity must be a positive whole number; odometer must be zero or greater; year must be plausible. Prevent duplicate registration numbers. Do not assign unavailable, maintenance-blocked or already allocated vehicles.
Driver Management	Validate required driver details, contact formats and licence information. Licence expiry must be a valid date; a driver with an expired licence cannot be assigned. Check NIC/licence formats where defined. Prevent overlapping driver assignments.
Maintenance Records	Require vehicle and maintenance details needed by the form, including service date and mileage where applicable. Mileage must be zero or greater. Next-service date/mileage must not precede the recorded service date/mileage. Vehicles marked unavailable for maintenance cannot be assigned.
Driver Trip Start / Completion	For the driver workflow in Chapter 1, require a valid starting odometer to start a trip. On completion, require an ending reading not lower than the starting reading. Calculate distance from the readings, do not allow the vehicle odometer to decrease, and limit remarks to the allowed field length.
Search, Status and Reports	For filters, validate dates and numeric ranges; start date must not be later than end date. Status values must be supported options. Show a helpful message when no records match. Display-only dashboards and lists need validation only if they include user input.

Validation Testing

- Submit each form with required fields empty; errors should appear and invalid data should not be saved.
- Try malformed email/phone values, mismatched passwords, invalid dates, and zero/negative counts or mileage.
- Test booking/assignment conflicts, insufficient vehicle capacity, maintenance blockouts and expired driver licences.
- Enter an ending odometer below the starting reading; trip completion should be blocked.
- Test success and failure outcomes for sandbox payment if implemented.
- Repeat key tests with valid data; entries should be accepted, stored and reflected in the relevant status/ confirmation.